MTH 3170 – Discrete Mathematics for CS

Set Theory (cont'd)

<u>Explain your reasoning on every problem. If a result is to be proven, then the proof should conform to the rules of proper grammar and punctuation.</u>

(1) If A = [0, 3], B = [2,7), with the universe being all real numbers, determine each of the following:

(a) $A \cap B$

(d) $A \cap \bar{B}$

(b) $A \cup B$

(e) $A - B$

(c) $\bar{A}$

(f) $B - A$

(2) Prove or disprove each of the following, given that sets *A, B, C,* and *D* are all subsets of universe $\mathcal{U}$.

<u>Extra Challenge:</u> If you are disproving a result, build the smallest counterexample you can build.

(a) If $A \subseteq B$ and $C \subseteq D$, then $A \cap C \subseteq B \cap D$.

(b) If $A \subseteq B$ and $C \subseteq D$, then $A \cup C \subseteq B \cup D$.

(c) $A \subseteq B$ if and only if $A \cap \bar{B} = \emptyset$.

(d) $A \subseteq B$ if and only if $\bar{A} \cup B = \mathcal{U}$.

(e) $A \cap C = B \cap C \Rightarrow A = B$.

(f) $A \cup C = B \cup C \Rightarrow A = B$.

(g) $A - (B \cup C) = (A - B) \cap (A - C)$.